THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ionic compound formed between a metal M with valency 3 and the sulphate ion has the formula: 1

(a) $M_2(SO_4)_3$ **(b)** MSO_4 **(c)** M_3SO_4 **(d)** $M(SO_4)_3$

2. Assertion (A): The formula of aluminium oxide is Al_2O_3. Reason (R): The total positive charge equals the total negative charge in the formula. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The total number of electrons transferred when 3 atoms of Mg combine with 2 atoms of N to form Mg_3N_2 is: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 2

4. The ionic compound between a metal of valency 2 and the phosphate ion has the formula: 1

(a) $M_2(PO_4)_3$ **(b)** MPO_4 **(c)** $M_3(PO_4)_2$ **(d)** M_2PO_4

5. Assertion (A): Water has a definite composition by mass. Reason (R): Hydrogen and oxygen in water are always in the ratio 1 : 8. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Which species has the same electron arrangement as neon? 1

(a) K^+ **(b)** Mg **(c)** Mg^{2+} **(d)** Cl^-

7. A solid X is hard, brittle, has a high melting point and conducts only when molten. X is: 1

(a) an ionic compound **(b)** a metal
(c) a covalent molecular compound **(d)** a non-metal element

8. Assertion (A): The atomic mass unit is defined with respect to carbon-12. Reason (R): One u is 1/12 of the mass of an atom of carbon-12. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** The atomicity of ozone is 3. 1
Reason (R): A molecule of ozone is made up of three oxygen atoms.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The molecular mass of water is 18 u. 1
Reason (R): The atomic mass of an element is expressed in the unified atomic mass unit (u).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is Fe_2O_3 and not FeO_3 the formula of rust-forming iron oxide? 2

12. Explain how the existence of isotopes modifies Dalton's postulate on the identical masses of atoms of an element. 2

13. The mass of an atom of oxygen compared with one-twelfth the mass of carbon-12 is 16. What is the relative atomic mass and unit? 2

14. State why metals form cations and non-metals form anions. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the percentage of nitrogen in 3
(a) NH_3
(b) NH_4NO_3 (N = 14, H = 1, O = 16) and say which is a better nitrogen fertiliser by percentage.

16. An element Y has the electron arrangement 2, 8, 7 and element Z has 2, 8, 8, 2. Predict the formula and bond in the compound they form. 3

17. Explain how formulae like Na_2CO_3 can show that carbonate is a unit and that the sum of charges is zero. 3

OR

Explain the properties of a covalent network solid like diamond and compare with a molecular solid like iodine. 3

18. Explain why carbon forms four bonds and shows covalent bonding with compounds like methane. 3

OR

The mass of an atom of element A is 12 times that of hydrogen and element B is 16 times. A compound AB2 contains how many parts by mass of A and B, and the molecular mass? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A farmer in Punjab reads three labels: calcium carbonate $CaCO_3$ (liming agent), sodium chloride $NaCl$, and ammonium sulphate $(NH_4)_2SO_4$ (fertiliser). Atomic masses: H 1, C 12, N 14, O 16, S 32, Ca 40. 5
(i) Find the formula unit mass of $CaCO_3$.
(ii) How many atoms are present in one formula unit of $(NH_4)_2SO_4$?
(iii) Find the formula unit mass of $(NH_4)_2SO_4$.
(iv) Find the percentage of nitrogen in $(NH_4)_2SO_4$ (to one decimal place).

20. Compute the molecular masses and the percentage of hydrogen in methane, ethane, ethene and ethyne and explain the trend (C = 12, H = 1). 5

OR

Plan an investigation to show that the formula of an oxide of copper is CuO: the procedure, the data to be collected, the calculations and the conclusion. 5

END OF QUESTION PAPER — Total: 40 Marks